Lecture 19 // Oct. 27 // SU(2) - Character Varieties

This week, we'll return to our discussion of the SU(2) character variety. We briefly mentioned this perspective when we discussed the critical points of the CS functional, but we haven't done any examples nor explicit calculations yet. Recall that we can study the fund. gp of a 3-mfd by considering instead the image of $\pi_1(Y)$ under some homomorphism to a simpler group, SU(2).

Def The Character Variety of a 3-mfd is

$$R(Y) = \left\{ \begin{array}{c} SU(2) \text{ representations} \\ \text{of } \pi_1(Y) \end{array} \right\} / \sim$$

where $\rho_0 \sim \rho_1$ IFF $\rho_1 = g^{-1} \cdot \rho_0 \cdot g$

Today, we'll discuss F-S's computation of IFH of Brieskorn spheres $\Sigma(a_1, \dots, a_n)$. Since we mentioned that

may need to perturb, will do

Recall that if the set of cp's of $CS(Y)$ is not discrete, we need to perturb the functional to get a discrete set.

With this in mind, we'll do ex's of both situations: when when R is discrete and when R has higher-dim components.

On Thursday, we'll discuss the common idea of viewing one's favorite gauge theory as the categorification of something. For IFHs, this "something" is the Casson invariant:

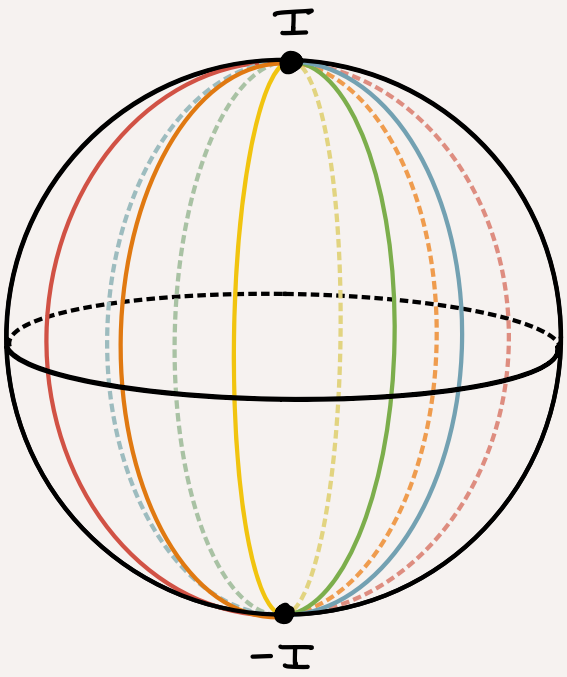
$$\frac{1}{2}\chi(H_*(CC(Y))) = \lambda(Y)$$

Examples of SU-2 char. varieties

First, lets recall some facts about $SU(2)$.

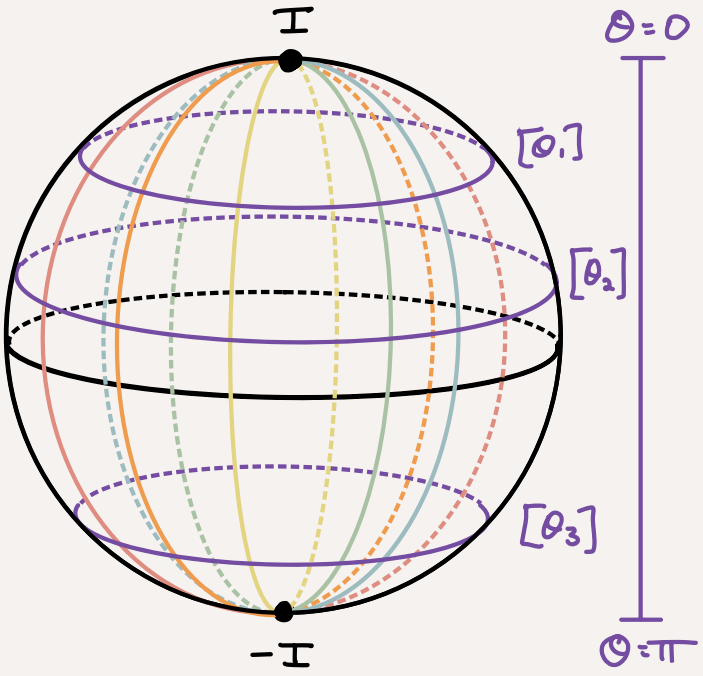
1) $SU(2) = \left\{ \begin{pmatrix} \alpha & -\overline{\beta} \\ \beta & \overline{\alpha} \end{pmatrix} \mid \alpha, \beta \in \mathbb{C},\ |\alpha|^2 + |\beta|^2 = 1 \right\}$ is diffeomorphic to S^3.

Going forward, we'll think of $SU(2)$ as S^3, which we'll draw 1-dimension down as S^2.



2) $SU(2)$ has many S^1 subgps, which are all conj to each other & sweep out $SU(2)$. They are all disjoint except at $\pm I$. Ie, every element of $SU(2)$ is conjugate to some diagonal matrix $\begin{bmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{bmatrix}$ which is unique up to the sign of θ.

3) For $A \in SU(2)$, $Tr(A)$ is a complete invariant up to conjugation: If A and B are conjugate, $Tr(A) = Tr(B)$.



Each conj class is a 2-sphere S^2 parameterized by the value of tr. For $\theta \in [0, \pi]$, we will speak of the conj. class of $[\theta]$ to mean the conj class of $\begin{bmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{bmatrix}$, where

$$\theta = \cos^{-1}\left(\frac{1}{2} Tr\left(\begin{bmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{bmatrix} \right) \right)$$

We can now compute the representation variety of a Brieskorn sphere $\Sigma(a_1,\dots,a_n)$, where the a_i are coprime.

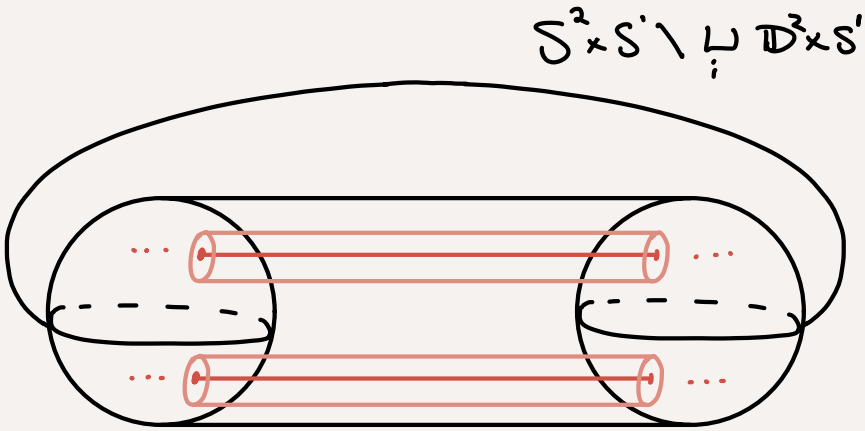
Def The Brieskorn spheres $\Sigma(a_1,\dots,a_n)$ are $\mathbb{Z}HS^3$'s constructed as follows:

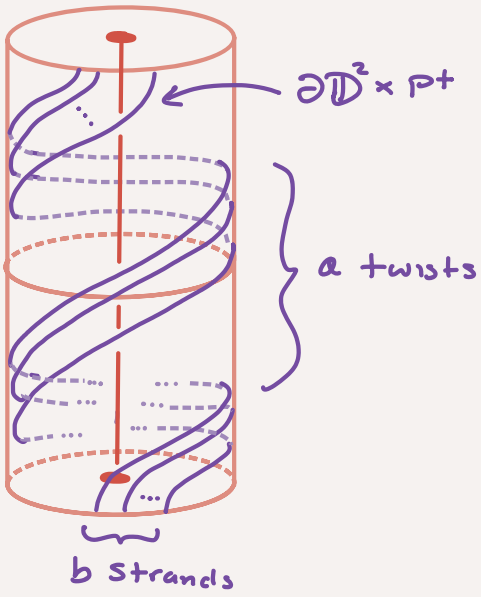
1) Solve the linear eq^n $\left(\prod_{i=1}^n a_i\right)\left(\sum_{i=1}^n \frac{b_i}{a_i}\right) = 1$

for a set $\{b_1,\dots,b_n\} \in \mathbb{Z}^n$

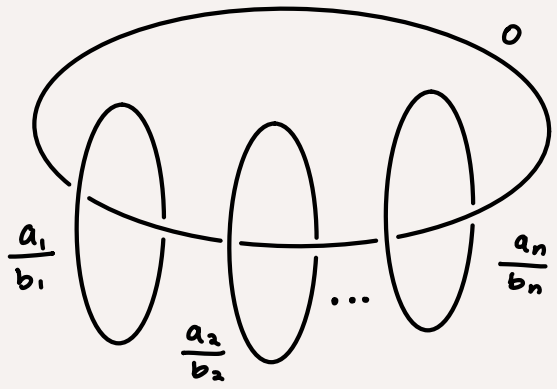
2) Surger $S^2\times S^1$ along n longitudinal fibres with surgery coefficients $\left\{\frac{a_i}{b_i}\right\}_{i=1}^n$.

Eg. punch out n longitudinal S^1 fibres to get n $S^1\times S^1$ boundary components. Let $pt\times S^1$ in $S^1\times S^2$ be homologous to $pt\times S^1$ in $S^2\times S^1$, and a/b be the class $a[S^1\times pt] + b[pt\times S^1]$. To each component, reglue $S^1\times D^2$ so that $[\partial D^2] \mapsto a[S^1\times pt] + b[pt\times S^1]$.

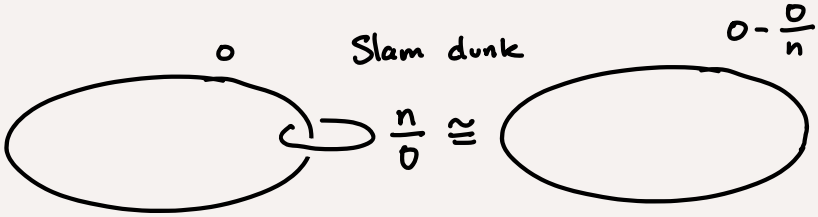




This gives the following surgery description



Rmk Sanity Check for coeffs.



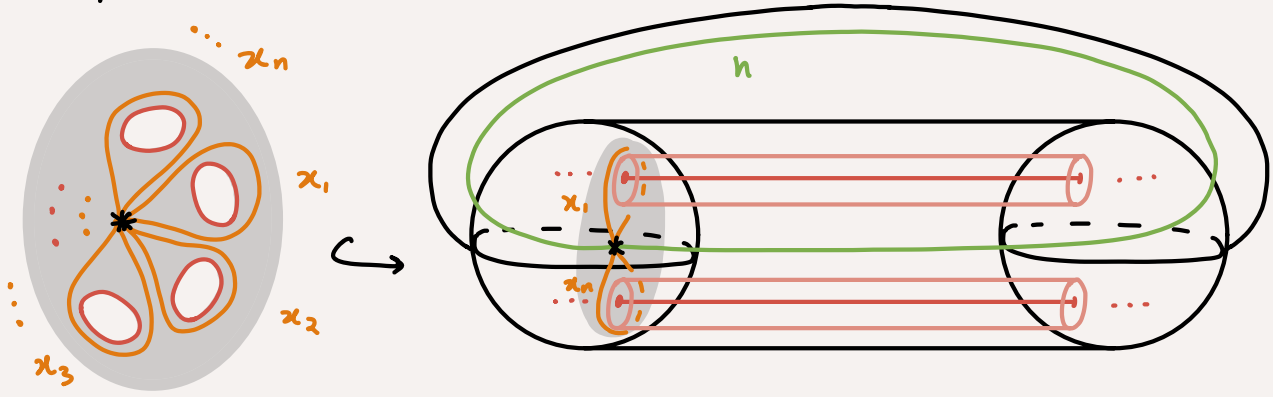
$\cong S^2 \times S^1$, so

$n \to S^1 \times pt \subset S^2 \times pt$

$0 \to pt \times S^1 \subset pt \times S^1$

Of course, the $\{a_1, \dots, a_n\}$ do not uniquely determine a 3-mfld. However, if we demand that the 3-mfld is a $\mathbb{Z}HS^3$, solving the equⁿ above gives a set $\{b_i\}_{i=1}^n$, which is not necc. unique, but one can show that they define the same 3-mfld. (This notion of equivalency can be ported over to equivalency of the sets $\{b_i\}$.

To get the Char var, we first must compute the fund gp. The generators are $x_1, \dots, x_n, h$, where the $x_1, \dots, x_n$ are loops about the meridional gen of each fibre and h is the longitudinal direction.



To get the relations, check that

- the longitude commutes with w/ each meridional gen., so $[x_i, h] = 1$ for all $1 \leq i \leq n$
- For each solid torus $\delta^1 \times \mathbb{D}^2$ we glue back in, $pt \times \partial\mathbb{D}^2$ gets attached along the (p,q) torus knot, so
$$[\partial \mathbb{D}^2] = x_i^{a_i} h^{b_i} = 1$$
- the product of all x_i's is nullhomotopic, so $\prod_{i=1}^{n} x_i = 1$

This gives the presentation for π_1:

$$\pi_1\left(\Sigma(a_1, \dots, a_n)\right) \cong \left\langle x_1, \dots, x_n, h \mid [x_i, h] = 1,\ x_i^{a_i} h^{b_i} = 1, \prod_{i=1}^{n} x_i = 1 \right\rangle$$

Now, lets think about the char var of this fund q_p.

To each gen, we need to assign an element of $SL(2)$ and this assignment must preserve the relations

We want to find irred rep^{n}s, eg, rep^{n}s with the smallest possible Stab under Conj action.

If a rep has a large Stab, then its trivial, since for ρ reducible, it factors thru the abelianization which is zero for homology spheres.

Lemma If ρ is irred then $\rho(h) = \pm Id$

Pf Assume not: $\rho(h) \neq \pm id$.

Then up to conjugation, $\rho(h)$ is diag matrix:

$$\rho(h) = \begin{bmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{bmatrix}, \quad \theta \neq 0, \pi$$

One can check that the only elements in $SU(2)$ which commute w/ a diag matrix are the elem. in that S^1 subgp:

if $$\begin{bmatrix} \alpha & -\bar{\beta} \\ \beta & \bar{\alpha} \end{bmatrix} \begin{bmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{bmatrix} = \begin{bmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{bmatrix} \begin{bmatrix} \alpha & -\bar{\beta} \\ \beta & \bar{\alpha} \end{bmatrix},$$

$$\begin{bmatrix} \alpha e^{i\theta} & -\bar{\beta} e^{-i\theta} \\ \beta e^{i\theta} & \bar{\alpha} e^{-i\theta} \end{bmatrix} = \begin{bmatrix} \alpha e^{i\theta} & -\bar{\beta} e^{i\theta} \\ \beta e^{-i\theta} & \bar{\alpha} e^{-i\theta} \end{bmatrix}$$

so $\beta e^{i\theta} = \beta e^{-i\theta}$ and $-\bar{\beta} e^{i\theta} = -\bar{\beta} e^{-i\theta}$,

so $\beta = 0$. Thus $\begin{bmatrix} \alpha & -\bar{\beta} \\ \beta & \bar{\alpha} \end{bmatrix} = \begin{bmatrix} \alpha & 0 \\ 0 & \bar{\alpha} \end{bmatrix}$

Since x_i commutes w/ h, this implies that, $\rho(x_i)$ is in the same subgp as $\rho(h)$, impling that ρ is reducible. □

So we may as well assume that $\rho(h) = \pm I$.

Now, lets ask ourselves what $\rho(x_i)$ can be for each i.

For each of the 2 choices, $\rho(h) = I$ or $\rho(h) = -I$.

using the third rel'n, we have that

$$\rho(x_i^{a_i} h^{b_i}) = \rho(1)$$

$$\rho(x_i)^{a_i} \rho(h)^{b_i} = 1$$

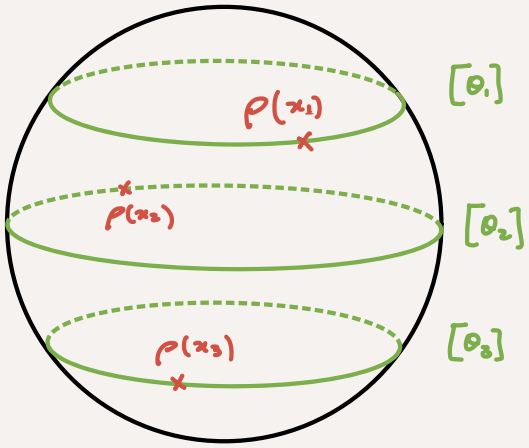
$$\rho(x_i)^{a_i} (\pm 1)^{b_i} = 1$$

$$\rho(x_i)^{a_i} = (\pm 1)^{-b_i}$$

Therefore $\rho(x_i)$ is a root of unity, so there is a discrete set of choices for the conj class of $\rho(x_i)$. In particular, $\rho(x_i)$ is in the conj class of a $\begin{cases} a_i^{th} \text{ root of unity, if } (\pm 1)^{-b_i} = id \\ 2a_i^{th} \text{ root of unity, if } (\pm 1)^{-b_i} = -id. \end{cases}$

Having fixed the (a_i, b_i)'s, for each tuple of choices for the conj. classes of the $\rho(x_i)$, we only need to enforce the last relation: $\pi \rho(x_i) = Id$

Another way of phrasing this constraint is: Fix conj classes $[\theta_i]$ for each x_i. Can we find $\rho(x_i) \in [\theta_i]$ such that $\rho(x_1) \cdot \ldots \cdot \rho(x_n) = Id$?

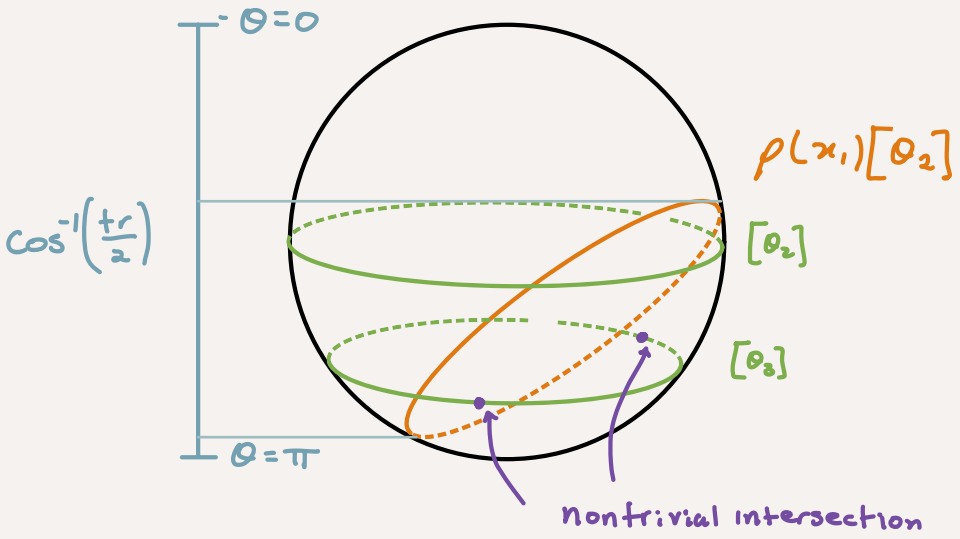


Lets now specialize to the case of $n=3$ singular fibres, $R(\Sigma(a_1, a_2, a_3))$. The last rel^n becomes $x_1 x_2 = x_3^{-1}$.

Since $\rho(x_3^{-1})$ is in the same cc as $\rho(x_3)$, $\rho(x_3)$ is in the cc as $\rho(x_1 x_2)$.

Up to conj, we can assume that $\rho(x_1)$ is in the std S^1 subgp, eg, is diag.

Then, we can ask if $\rho(x_1)\cdot[\theta_2] \cap [\theta_3] \neq \emptyset$?



Since left mult. by a fixed element of $SU(2)$ is an isometry of $SU(2)$, $\rho(x_1)[\theta_2]$ is another S^2 but it is slanted.

To check whether it intersects with $[\theta_3]$, we can look at the range of values of $\cos^{-1}(\frac{1}{2}tr)$ on the S^2 given by $\rho(x_1)[\theta_2]$.

Since the norm of matrices in $SU(2)$ is 1, the trace is is minimized/maximized when the matrix is diagonal, so the endpoints must be at $|\theta_1-\theta_2|$ and $\theta_1+\theta_2$ (or, if $\theta_1+\theta_2>\pi$, $2\pi-(\theta_1+\theta_2)$).

$$\begin{bmatrix} e^{i\theta_1} & 0 \\ 0 & e^{-i\theta_1} \end{bmatrix} \cdot \begin{bmatrix} e^{i\theta_2} & 0 \\ 0 & e^{i\theta_2} \end{bmatrix} = \begin{bmatrix} e^{i(\theta_1-\theta_2)} & 0 \\ 0 & e^{i(\theta_1-\theta_2)} \end{bmatrix}$$

$$\begin{bmatrix} e^{i\theta_1} & 0 \\ 0 & e^{-i\theta_1} \end{bmatrix} \quad \begin{bmatrix} e^{i\theta_2} & 0 \\ 0 & e^{-i\theta_2} \end{bmatrix} \quad \begin{bmatrix} e^{i(\theta_1+\theta_2)} & 0 \\ 0 & e^{-i(\theta_1+\theta_2)} \end{bmatrix}$$

Therefore $\cos^{-1}\left(\frac{tr(A)}{2}\right)$ when $A \in \rho(x_1)[\theta_2]$. lies in the interval

$$\left[|\theta_1 - \theta_2| \,,\; \begin{cases} \theta_1 + \theta_2 & \text{if } \theta_1 + \theta_2 \in \pi \\ 2\pi - (\theta_1 + \theta_2) & \theta_1 + \theta_2 > \pi \end{cases} \right]$$

If θ_3 is not in this interval, then we don't get a rep^n for that choice of conj classes. If it does, then in general, the intersection $\rho(x_1)\cdot[\theta_2] \cap [\theta_3]$ will be a copy of S^1, since generically $S^1 = S^2 \cap S^1$ in S^3.

But remember that we're quotienting by conjugation! We alv. quot. a bit by choosing θ_1 in the skel S^1, but we can still conjugate within this S^1 & preserve the intersection. This S^1 is the "leftover" conj by the S^1 subgp which contains $\rho(x_1)$. After doing so, for each $\{\theta_1, \theta_2, \theta_3\}$, we get one pt in the char var.

Rmk If the int is in a pt (instead of an S^1), this says that the leftover action fixes a pt, which implies that the x_i all commute, so π_1 is abelian and hence reducible.

Rmk Since the a_i are coprime, θ_3 can never be at an end of the interval.

Lets do an example for our favorite 3-mfd so far: $\Sigma(2,3,5)$,

Step 1: Solve the eqn $\left(\prod_{i=1}^{3} a_i\right)\left(\sum_{i=1}^{3} \frac{b_i}{a_i}\right) = 1$:

$$(2\cdot 3\cdot 5)\left(\frac{b_1}{2} + \frac{b_2}{3} + \frac{b_3}{5}\right) = 1$$

$$15b_1 + 10b_2 + 6b_3 = 1$$

Notice: $15+6=21$, $10\cdot 2=20$, $21-20=1$

$\Rightarrow$ $b_1=b_3=-1$, $b_2=2$ works

Step 2: Compute π_1:

$$\pi_1\left[\Sigma(2,3,5)\right] = \langle x_1, x_2, x_3, h \mid [x_i,h]=1,\ x_1^2=h,\ x_2^3=h^{-1},\ x_3^5=h^{-1},\ x_1x_2x_3=1\rangle$$

Step 3: Find rep's which satisfy the first two rel'ns:

If $\rho(h)=I$:

$\rho(x_1)^2 = Id \Rightarrow \rho(x_1) = \pm Id$

$\Rightarrow \rho(x_2x_3)=1 \Rightarrow \rho(x_2)\rho(x_3)=\pm Id$

$\Rightarrow$ rep^n is reducible

If $\rho(h)=-Id$:

$\rho(x_1)^2 = -Id \Rightarrow \rho(x_1)^4 = Id$

$\Rightarrow \rho(x_1)$ is a 4th root of unity whose square is -1

$\Rightarrow \rho(x_1) \in [\pi/2]$

$\rho(x_2)^3 = -(-Id) = Id \Rightarrow \rho(x_2) \in [\pi/3]$ or $[2\pi/3]$

$\rho(x_3)^5 = -(-Id) = Id \Rightarrow \rho(x_3) \in [n\pi/5]$, $n=1,2,3,4$

Step 4: Use the range of $\rho(x_1)[\mathbb{Q}_2]$ to verify the last relation:

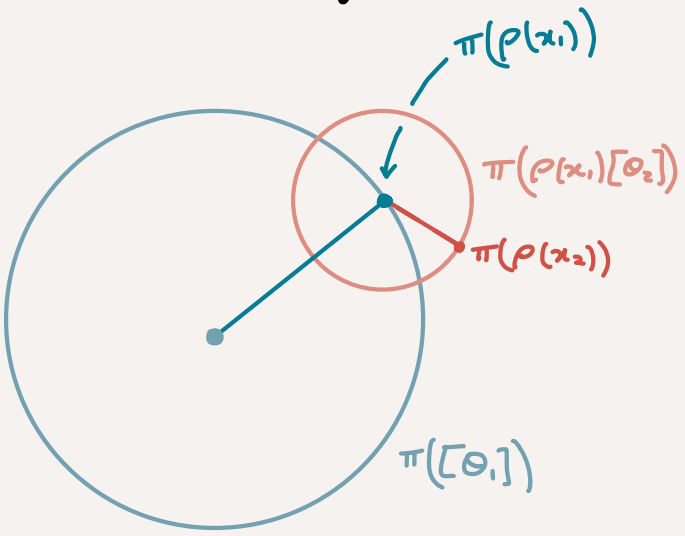
INSERT THIS STEP

It turns out that in general, the Char-Var for $n=3$ is a discrete set. Strictly speaking, this is still not enough to prove that no perturbation is needed (we still need to verify transversality, like in normal morse theory) but in this case, everything works out. But it is true that if the Char-Var is not a discrete set, we definitely must perturb. Lets now take $n=4$, and show how $R(\Sigma(a_1,a_2,a_3,a_4))$ need not be discrete.

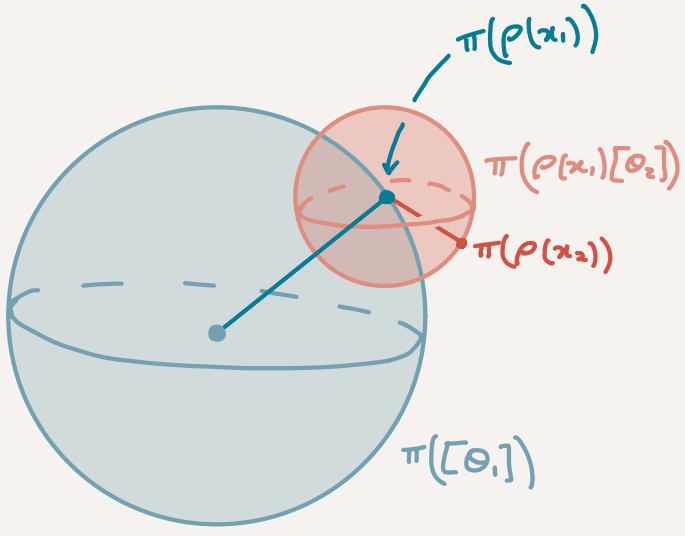
Here is the same story up until we get to the third condition. For each 4-tuple of conj classes $\{[\Theta_1],[\Theta_2],[\Theta_3],[\Theta_4]\}$, we ask if we can choose representatives in each class satisfying $x_1x_2x_3x_4=1 \Leftrightarrow x_1x_2 = x_4^{-1}x_3^{-1}$.

As before, we formulate this as finding $\rho(x_1) \in [\Theta_1]$ and $\rho(x_4^{-1}) \in [\Theta_4]$ such that $\rho(x_1)[\Theta_2] \cap \rho(x_4^{-1})[\Theta_3] \neq \emptyset$.

To better visualize this condition, lets look at the image of Stereographic proj $\pi: S^3 \to \mathbb{R}^3$. Since π is conformal, spheres get mapped to spheres. The size of each $[\Theta_i]$ sphere is inversely proportional to Θ_i, and rotating $[\Theta_2]$ by $\rho(x_1)$ corresponds to translating the center of $\pi([\Theta_2])$ to lie at $\rho(x) \in \mathbb{R}^3$

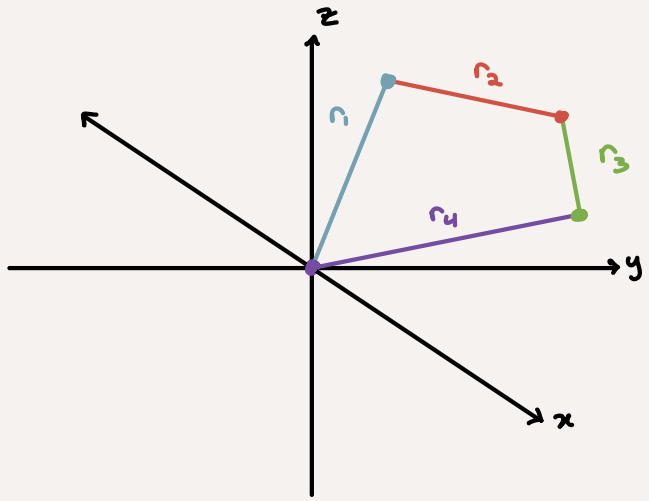


2-d picture



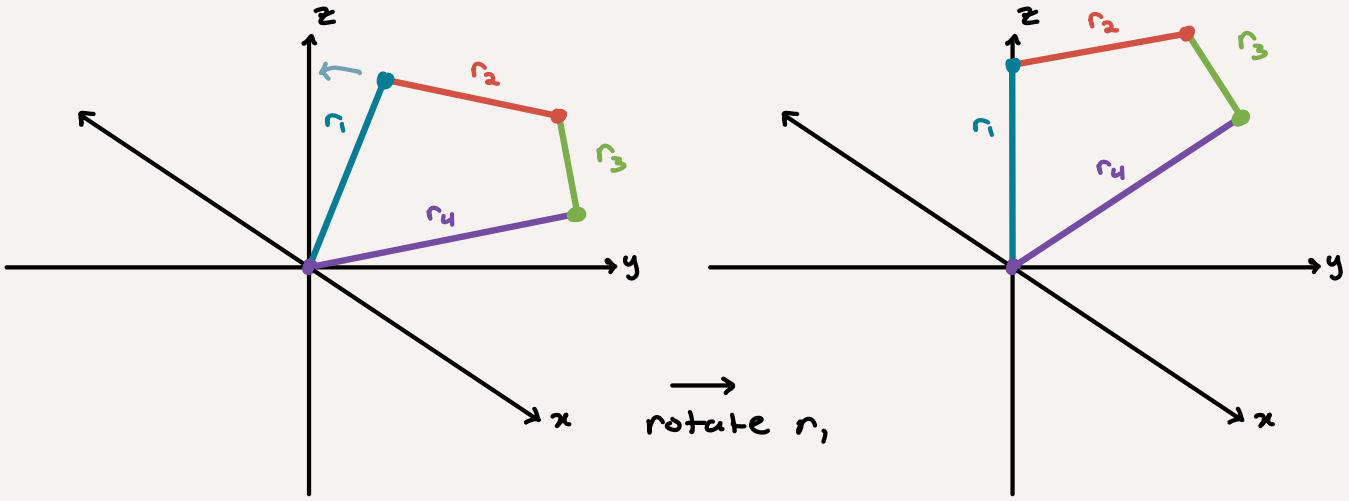
3-d picture

However, trying to include the other two spheres makes this picture very busy! The relevant data is the radius of each sphere, and the location of the center-point of the subsequent sphere. So an even better way to visualize the condition that $x_1x_2x_3x_4 = 1$ is by a collection of four rigid rods whose lengths depend on θ_i, where the start of rod i is attached to the end of rod $i-1$. Then $x_1x_2x_3x_4 = 1$ exactly when the rods form a closed polygon in $\mathbb{R}^3$.

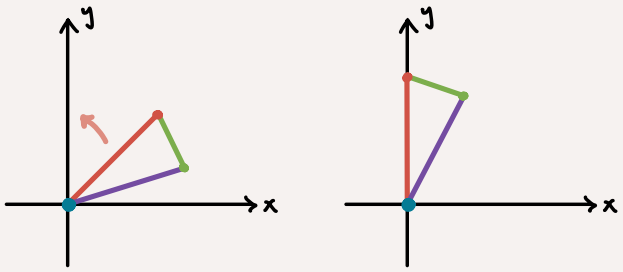


In general, such a configuration of rods for a given set of side lengths does exist. The question then becomes how many are there, up to conjugation. The conjugation action is that of $SU(2)$ acting on $\mathbb{R}^3$, so by rotations.

We may first move r_1 to align w/ the z axis. This is the same thing as conjugating to turn $\rho(x_1)$ into a diagonal matrix. The leftover action corr. to rotating about the z-axis, since that fixes $\rho(x_1)$. Therefore we can rotate such that r_2 lies in the RHS of the y-z plane. However, there's still some deg's of freedom! so the moduli space is still 1-dimensional.

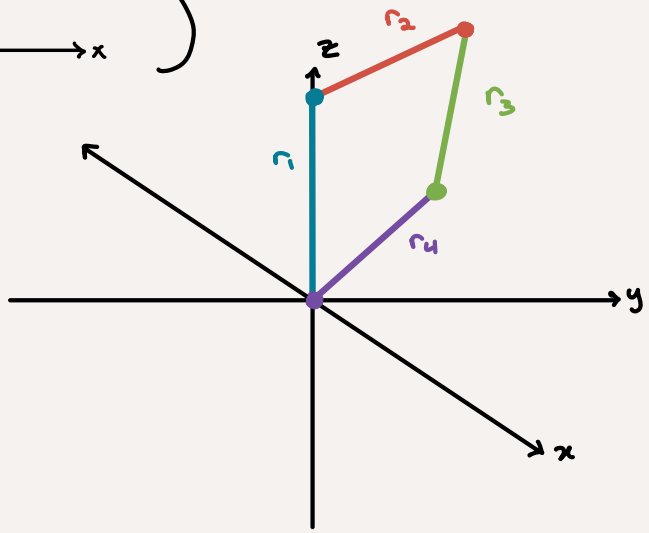


Top down r_2 rotation:

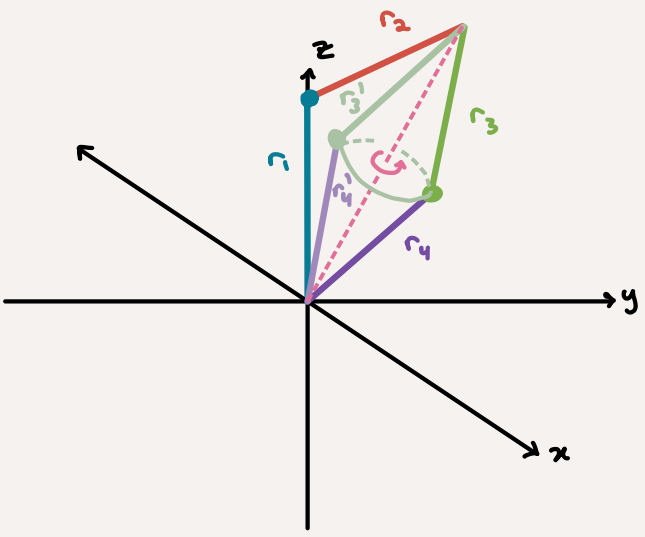


rotate r_2

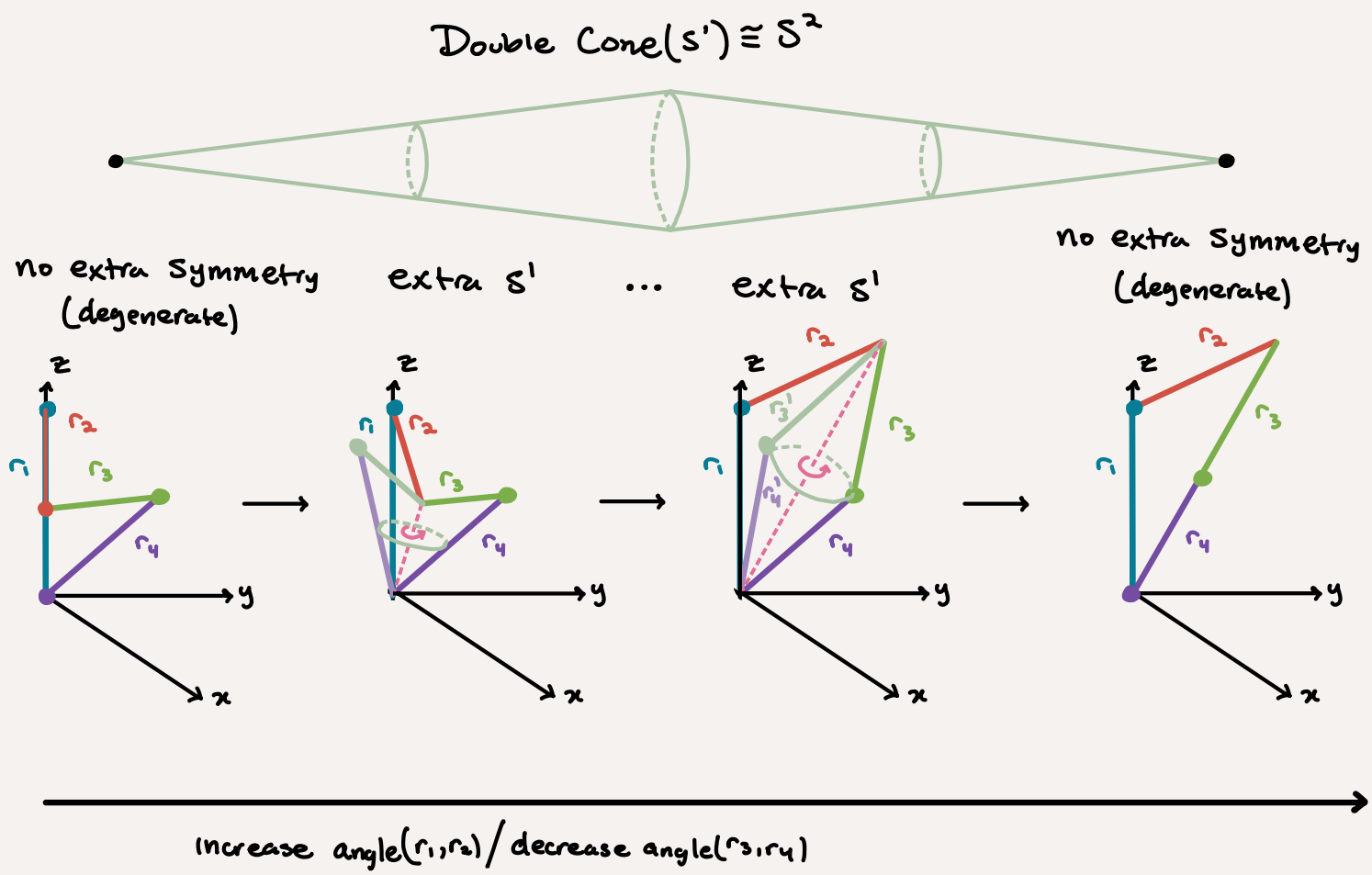
This is the final polytope →



However, there is still an S^1 action on this polytope which is not given by conjugation! In fact this S^1 action is a slice of a residual S^2's worth of action in $\mathbb{R}$.



This S^2 can be thought of as the double cone on S^1, where the cone points are the degeneration of the symmetry due to the angle between r_1 and r_2, and r_3 and r_4.


Double Cone(S') ≅ S²
no extra Symmetry
(degenerate)
extra S¹
...
extra S¹
no extra Symmetry
(degenerate)
z
x
y
r1
r2
r3
r4
Increase angle(r1,r2) / decrease angle(r3,r4)